29/11/18

Indian Institute of Engineering Science & Technology, Shibpur

B.Tech 3^{rd} Semester (CST) Examination, 2018

Digital Logic (CS 301)

Answer any 5 questions.

F.M. 70

Time: 3 hrs

1. (a) Simplify the following Boolean function to minimum number of literals

$$y(w\overline{z} + wz) + xy.$$

(b) Express the Boolean function $F = xy + \overline{x}z$ as a product of maxterms.

(c) Draw the voltage transfer characteristic of an inverter and define the following parameters: V_{OH}, V_{OL}, V_{IL}, and V_{IH}. Obtain the expression for noise margins from the said parameters.

[3 + 4 + 7]

2. (a) Why wired-logic connection is not allowed with totem-pole (TTL) output circuit?

(b) Which logic level limits the fan-out of DTL gates and why?

(c) Draw the circuit diagram of a two inputs DTL NAND gate. Connect the output of the DTL gate to N inputs of other similar gates. Assume that the output transistor is saturated and its base current is 0.44mA. Let h_{FE} of the transistor be 20. Find the value of N that will keep the transistor in saturation and what is the fan-out of the gate?.

[4 + 3 + 7]

3. (a)Using four XOR gates and minimum number of full-adders, construct a 4-bit parallel adder/subtractor circuit. Use an input select variable S so that when S = 0, the circuit adds and when S = 1, the circuit subtracts.

(b) Design a combinational circuit that will act as 4-bit binary to gray code converter.

(c) Draw the circuit diagram of a tri-state TTL inverter.

[5 + 5 + 4]

4. (a) Design a four (three message bits and one parity bit) bit even parity-generator-cum-checker circuit and also draw the logic diagram of the circuit.

(b) An 8 × 1 multiplexer has inputs A, B, and C connected to the selection inputs S_2, S_1 and S_0 respectively. The data inputs I_0 through I_7 are as follows:
$I_1 = I_2 = I_7 = 0$; $I_3 = I_5 = 1$; $I_0 = I_4 = D$ and $I_6 = \overline{D}$. Determine the Boolean function that the multiplexer implements .

[7 + 7]

5. (a) Draw the circuit diagram of a mod-16 ripple counter and also draw the waveforms (timing diagram) of the said counter.

b) Define state equations of sequential circuits? Design a sequential circuit with JK flip-flops to satisfy the following state equations:

$$A(t+1) = xAB + y\overline{A}C + xy$$
$$B(t+1) = xAC + \overline{y}B\overline{C}$$
$$C(t+1) = \overline{x}B + yA\overline{B}$$

[6 + 8]

6. (a) Draw the block diagram of a 4-bit bi-directional shift register and explain its operation.

(b) Design a counter with the following binary sequence: 0, 1, 3, 2, 6, 4, 5, 7 and repeat. Use RS flip-flops in the design process. [7 + 7]

7. Define lockout of counter. Design a mod-5 counter using JK flip-flop so that if the unused states 101, 110 and 111 occur, the next clock pulse will reset the counter to 000. [3 + 11]